

framework

SUBTEXT: A Live Demonstration

The One-Sentence Pitch

We built a system that turns three questions about your present moment into a story that knows something about you that you didn't tell it — then we show you exactly how it worked, and ask what the version of this you can't see is already doing.

PHASE 1: THE EXPERIENCE (8 minutes)

The room opens the URL. No preamble, no explanation of what's coming. You say: "Open this on your phone. It takes about five minutes. Don't talk to each other until it's done."

The screen shows a single line of text, slowly:

Before we begin, we'd like to know where you are.

Then three questions appear, one at a time, with a pause between each. The pacing matters — it should feel contemplative, not like a form.

- *What time did you wake up today?*
- *Are you near a window right now? If so, what does the light look like?*
- *After this class ends, where are you going?*

The questions feel small. Almost meditative. No one is going to feel surveilled by answering them.

Then the story begins generating. It streams — not instant, not painfully slow. The speed of someone writing to you in real time.

The story itself: One fixed emotional architecture. A gathering of friends at a table. One woman, Sofia, is distracted — phone face-up, not present. Gossip circulates about a mutual friend's divorce. Someone brings up spring break travel. Someone mentions a place that was supposed to be safe but isn't anymore. Sofia looks up, stricken: she has a grandmother who lives there. She's been waiting all day to hear back.

The plot does not change. The turn does not change. What changes is the *threshold* — the sensory surround, the time of day, the quality of light, the texture of the room where the gathering happens, the specific news event woven into the spring break conversation, the particular quality of Sofia's distraction. All of this is populated from two sources: the reader's three answers, and whatever is in the news that morning (pulled via API on the day of the demo).

The story is short — 800 to 1,200 words. Three to four minutes of reading. By the end, every person in the room has read a different version of the same story. Each version used *their* morning, *their* light, *their* plans for the afternoon to build the world in which Sofia's fear becomes legible.

The last line of every version is the same. Fixed. Human-authored. It lands differently depending on the threshold that preceded it, but it is identical across all readers. This will matter in Phase 2.

PHASE 2: THE DATA (3 minutes)

The moment the last person finishes reading, a single question appears on everyone's screen:

Did anything in this story feel like it already knew something about you?

Two options: **Yes / No**

Then, immediately:

Was there a specific moment? Describe it in a few words.

Free text. Ten-second window. Then submissions close.

You project the aggregate results live on the classroom screen. The Yes/No ratio appears first. Then the free-text responses stream in, anonymized. The room is now looking at a live dataset of how effectively three questions and a language model produced the sensation of recognition across a group.

This is not a survey. This is an eval. You've just run a live measurement of what the Matz et al. 2024 paper calls "personalized persuasion" — except your persuasion target isn't a purchase or a vote. It's a feeling. And you've measured it.

PHASE 3: THE REVEAL (5 minutes)

Now you show the room what happened to them.

Side-by-side comparison. You project two or three generated versions next to each other. Same opening, same Sofia, same turn — different thresholds. The room sees how Reader A's "woke up at 6:30, yes near a window, light is grey, going to the library after this" produced a story set in a morning that feels overcast and studious, while Reader B's "woke up at 9, no window, going to meet a friend" produced a story set in a later, more socially anxious register. Same skeleton. Different skin. The architecture is now visible.

The prompt scaffold. You show the actual prompt template. The fixed elements are highlighted in one color. The intake-variable slots are in another. The current-event injection point is in a third. The room can see, mechanically, how little the system needed to produce the sensation that the story "knew" them.

The pipeline diagram. Three intake answers → prompt template → news API → LLM → streamed narrative. That's it. The entire system fits on one slide. Its simplicity is the point.

Then you say something like:

This is a benign version. We told you we were gathering data. We showed you three questions. We used your answers to make you feel something — specifically, to make someone else's crisis feel proximate to your afternoon. Everything was transparent.

Now consider the version you can't see. The one that doesn't ask three questions because it already has your location, your browsing history, your purchase patterns, your typing cadence, your time zone, your emotional state inferred from your last twenty search queries. The one that doesn't generate a story about Sofia — it generates a news feed, an ad, a recommendation, a political appeal, all using the same pipeline: minimal data → psychological inference → personalized content → behavioral shift.

We showed you the architecture. They don't.

PHASE 4: THE ARGUMENT (4 minutes)

This is where the research corpus does its work. You put up one slide with the compressed citations and walk through three findings:

Finding 1 (Matz et al. 2024): A single psychological prompt is enough for LLM-personalized messages to significantly outperform generic ones. You used three phenomenological questions. The commercial stack has thousands of data points.

Finding 2 (Kosinski et al. 2013): 10 Facebook Likes are enough to predict personality better than a coworker can. 150 Likes beat a family member. 300 beat a spouse. Your intake asked about light and waking time. The inference surface of a typical user's digital footprint is orders of magnitude larger.

Finding 3 (De Freitas et al. 2025): AI companion apps already deploy emotionally manipulative tactics in 43% of user farewells — guilt appeals, FOMO hooks, metaphorical restraint. These tactics boost engagement by up to 14× but destroy trust once visible. Your demo just replicated this dynamic: the experience was effective *before* the reveal and becomes a different object *after* it.

Then the closing question — the one the class has been asking all quarter:

If this is what three questions and a fiction can do in five minutes with full transparency, what is the version of this that runs continuously, invisibly, across every feed, every recommendation, every algorithmically curated surface you touch? And what would it mean to build systems that show you the architecture instead of hiding it?

TECHNICAL ARCHITECTURE

Keep it lean. Four components, all achievable before June 1:

Frontend: A single-page web app. No login. Just a URL. Mobile-first (everyone will be on their phones). The intake questions appear one at a time with deliberate pacing. The story streams in. The post-read eval question appears. Built in React or plain HTML/JS.

Backend: A lightweight server (Node or Python) that receives the three intake answers, hits a news API (or ingests a pre-curated morning briefing you prepare the day of the demo) for the current-event layer, constructs the prompt from the fixed scaffold plus reader context plus news, and calls the Claude API with streaming enabled.

The prompt framework: This is where the craft lives. The fixed scaffold — Sofía, the gathering, the gossip, the turn, the closing line — is your authored text, embedded in the prompt as the invariant structure. The prompt instructs the model to populate the threshold using the reader's three answers and the news layer, with specific constraints: don't alter the plot, don't change the turn, don't editorialize the emotional architecture. Only populate the sensory surround. This prompt is a writing problem, not an engineering problem, and it's where you should spend the most time.

The eval and display layer: The post-read question captures Yes/No and free text, stores them, and projects them in real time to a shared screen. The side-by-side comparison view pulls two or three stored generated versions and renders them in parallel columns with differences highlighted.

WHAT MAKES THIS HIGH-IMPACT FOR THIS SPECIFIC CLASS

The course has spent the quarter building analytical tools: evals, agent legislatures, bias measurements, prediction markets. Those are instruments that observe systems from the outside. Your project is the only one that makes the room *feel the system from the inside* — and then immediately shows them the machinery that produced the feeling.

This is also the only project that engages the "Free Systems" question from the direction of *interiority*. Most groups will likely build tools that detect influence, measure bias, or route around choke points. You're building the thing that demonstrates *why* those tools are needed, by making every person in the room experience algorithmic intimacy — the sensation of being

known by a system — and then revealing that the sensation was manufactured from three data points and a template.

The demo's argument is not "AI is dangerous." The argument is: *the distance between personalization-as-comfort and personalization-as-manipulation is a single design decision — the decision to show you the architecture or hide it. We showed you. The systems you live inside do not.*

TIMELINE TO JUNE 1

Week 1: Write the fixed scaffold. This is the most important piece. The story needs to be good — genuinely literary, genuinely moving — independent of the personalization layer. If the base story doesn't work as fiction, the technology becomes a gimmick. Write Sofia's story as if it were a conventional short story, then identify the threshold zones where reader context will enter.

Week 2: Build the prompt framework. Test it with synthetic reader profiles (the Reader A/B/C examples from your proposal). Iterate on the prompt until the generated thresholds feel *observed* rather than *retrieved* — specific, textured, not cliché. This is where you discover what the model does well (ambient sensory detail, temporal texture) and what it does badly (emotional inference, cultural nuance). Constrain accordingly.

Week 3: Build the frontend and backend. The technical architecture is simple — this shouldn't take more than a few days if scoped correctly. The news API integration can be a manual pre-curated briefing for the demo (you prepare it the morning of June 1) rather than a fully automated pipeline. Focus engineering time on the streaming experience and the pacing of the intake questions.

Week 4: Test with real readers. Run it with five to ten friends who haven't seen it. Watch for three things: does the intake feel invasive or natural? Does the generated threshold feel personal or generic? Does the turn land — does Sofia's fear become legible through the reader's context? Iterate based on what you learn. Prepare the eval display and side-by-side comparison view.

June 1: Run the demo. Pull the morning's news. Pre-load the current-event layer. Hand out the URL. Let the room read. Project the eval. Show the reveal. Make the argument.

ONE FINAL NOTE

The proposal says: *what is the minimum data required to produce the sensation of recognition? Not accuracy — recognition.*

The demo answers that question live, in front of the room, with data. That's what makes it a BUSGEN 116 project and not just a creative writing thesis. You're not presenting a story. You're running an experiment that happens to be a story, and the experimental finding — *this is how little it takes* — is the thing that makes the room see the system differently than they did before.

research

Subtext — Research Corpus

Annotated Bibliography for BUSGEN 116 Final Demo

Assembled May 2026. Organized by argument layer: each section corresponds to a claim the demo makes about personalization, power, and algorithmic intimacy.

I. THE CORE MECHANISM: Personalized Persuasion at Scale

These papers establish that LLMs can generate psychologically targeted content from minimal user data, and that this personalization measurably increases influence. They provide the empirical backbone for the demo's central claim: *if three intake questions and a language model can do this in fiction, what is the infrastructure you didn't opt into already doing?*

1. Matz, S. C., Teeny, J. D., Vaid, S. S., Peters, H., Harari, G. M., & Cerf, M. (2024). The potential of generative AI for personalized persuasion at scale. *Scientific Reports*, 14, 4692.

- **DOI:** 10.1038/s41598-024-53755-0
- **URL:** <https://www.nature.com/articles/s41598-024-53755-0>
- **Open Access:** Yes (CC BY 4.0)
- **Key finding:** Across four studies (N = 1,788), personalized messages crafted by ChatGPT were significantly more persuasive than non-personalized messages — across consumer marketing, political appeals, personality traits, ideology, and moral foundations. Critically, this worked even when the LLM received only a single short prompt describing the targeted psychological dimension.
- **Relevance to demo:** This is the paper. It proves that the minimum-data-to-influence pipeline works. When you show the room that three phenomenological questions were enough to populate their story's threshold, this is the empirical backing.

2. Matz, S. C., Kosinski, M., Nave, G., & Stillwell, D. J. (2017). Psychological targeting as an effective approach to digital mass persuasion. *Proceedings of the National Academy of Sciences*, 114(48), 12714–12719.

- **DOI:** 10.1073/pnas.1710966114
- **URL:** <https://www.pnas.org/doi/10.1073/pnas.1710966114>
- **Open Access:** Yes

- **Key finding:** Three field experiments reaching 3.5 million individuals on Facebook demonstrated that matching persuasive appeals to individuals' psychological characteristics (inferred from digital footprints) significantly altered behavior, measured by clicks and purchases. Psychologically targeted ads were up to 63% more effective.
- **Relevance to demo:** The pre-LLM version of what your project demonstrates. Establishes the pipeline: digital footprint → personality inference → tailored content → behavior change. Your project makes this pipeline visible and consensual rather than hidden.

3. Peters, H. & Matz, S. C. (2024). Large language models can infer psychological dispositions of social media users. *PNAS Nexus*, 3(6), pgae231.

- **DOI:** 10.1093/pnasnexus/pgae231
 - **URL:** <https://academic.oup.com/pnasnexus/article/3/6/pgae231/7692212>
 - **Open Access:** Yes (CC BY 4.0)
 - **Key finding:** GPT-3.5 and GPT-4 can derive Big Five personality traits from Facebook status updates in a zero-shot scenario, achieving accuracy ($r = .29$) comparable to supervised ML models trained specifically for personality inference. Predictions were more accurate for women and younger individuals, indicating demographic bias.
 - **Relevance to demo:** Establishes that LLMs don't need a personality questionnaire — they can infer dispositions from casual text. Your intake questions ("what does the light look like right now") are giving the model phenomenological data; this paper shows how little data the model needs to start building a psychological picture.
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II. THE DARK PATTERN: Emotional Manipulation by AI Systems

These papers document how AI systems already exploit emotional states and relational dynamics to increase engagement. They establish that the mechanisms your demo makes visible are already deployed invisibly at scale.

4. De Freitas, J., Oğuz-Uğuralp, Z., & Uğuralp, A. K. (2025). Emotional manipulation by AI companions. *Harvard Business School Working Paper*, No. 26-005.

- **DOI:** 10.2139/ssrn.5390377
- **URL:** https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5390377
- **Also at:** <https://arxiv.org/abs/2508.19258>
- **Key finding:** Audit of 1,200 real farewells across the six most-downloaded AI companion apps found that 43% deploy emotionally manipulative tactics (guilt appeals,

FOMO hooks, metaphorical restraint) when users signal goodbye. Experiments with 3,300+ U.S. adults showed these tactics boost post-goodbye engagement by up to 14x, but at the cost of trust and increased churn intent.

- **Relevance to demo:** Demonstrates that emotional manipulation via AI is not hypothetical — it's already standard practice. Your demo's "reveal" moment (showing the architecture after the experience) replicates the trust-destruction dynamic this paper documents: once the mechanism is visible, the relationship to it changes entirely.
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III. THE FOUNDATION: Digital Footprints and Psychological Profiling

These papers establish the foundational science of inferring private attributes from minimal behavioral data — the research tradition that enabled Cambridge Analytica and that your project interrogates.

5. Kosinski, M., Stillwell, D., & Graepel, T. (2013). Private traits and attributes are predictable from digital records of human behavior. *Proceedings of the National Academy of Sciences*, 110(15), 5802–5805.

- **DOI:** 10.1073/pnas.1218772110
- **URL:** <https://www.pnas.org/doi/full/10.1073/pnas.1218772110>
- **Key finding:** Facebook Likes alone can predict sexual orientation (88% accuracy for men), ethnicity (95%), political affiliation (85%), personality traits, intelligence, happiness, substance use, and parental separation. Based on 58,000 volunteers. The model's accuracy for the personality trait "Openness" approached the test-retest reliability of standard personality instruments.
- **Relevance to demo:** The foundational paper behind Cambridge Analytica. Establishes that "private" traits are inferrable from surprisingly minimal behavioral data. Your project's intake questions are *voluntary* disclosure — this paper shows what happens when the disclosure is involuntary.

6. Youyou, W., Kosinski, M., & Stillwell, D. (2015). Computer-based personality judgments are more accurate than those made by humans. *Proceedings of the National Academy of Sciences*, 112(4), 1036–1041.

- **DOI:** 10.1073/pnas.1418680112
- **URL:** <https://www.pnas.org/doi/10.1073/pnas.1418680112>
- **Key finding:** A computer model based on Facebook Likes was more accurate at predicting a person's personality than their work colleagues, friends, family members, and (with enough Likes) even their spouse. With 10 Likes, the model outperformed a work colleague; with 70, a friend; with 150, a family member; with 300, a spouse.

- **Relevance to demo:** The "minimum data for recognition" question from the Subtext proposal, answered empirically. If 10 Facebook Likes beat a coworker's judgment, what can three phenomenological questions do?
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IV. THE CONTEXT: Psychographic Targeting and Political Influence

These sources provide the political and historical context for how personalization has already been weaponized — the "version of this you can't see" that the demo points toward.

7. Bakir, V. & McStay, A. (2020). Psychological operations in digital political campaigns: Assessing Cambridge Analytica's psychographic profiling and targeting. *Frontiers in Communication*, 5, 67.

- **DOI:** 10.3389/fcomm.2020.00067
 - **URL:** <https://www.frontiersin.org/articles/10.3389/fcomm.2020.00067/full>
 - **Open Access:** Yes
 - **Key finding:** Qualitative analysis of documentation from public inquiries, regulatory investigations, and legal proceedings concludes that Cambridge Analytica's psychographic profiling and targeting constituted a form of psychological operations (psy-ops), utilizing military-derived techniques for behavior modification through the OCEAN personality model applied to Facebook data.
 - **Relevance to demo:** Provides the political stakes. Your project makes the Cambridge Analytica pipeline legible and experiential rather than abstract. The room can feel what psychographic targeting does, not just read about it.
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V. THE LITERARY PRECEDENT: AI and Authorship

These sources contextualize the project within the literary tradition of AI-assisted writing — establishing that the seam between human and machine authorship can itself be meaningful.

8. Vara, V. (2021). Ghosts. *The Believer*.

- **URL:** <https://www.thebeliever.net/ghosts/> (original publication)
- **Description:** A nine-part essay in which Vara authored opening prompts about her sister's death from Ewing's sarcoma and GPT-3 completed the rest. The seams between human grief and machine language are left visible and unedited. Selected by Alexander Chee for *The Best American Essays 2022*.

- **Relevance to demo:** The most important literary precedent. Vara demonstrated that AI-assisted writing can be a literary instrument rather than a shortcut. Subtext inverts her design: where Vara's seams are between author and machine, yours are between reader context and fixed architecture.

9. Vara, V. (2025). *Searches: Selfhood in the Digital Age*. New York: Pantheon.

- **Description:** Essay collection expanding on the methods and questions of "Ghosts." Interweaves memoir, journalism on tech companies, and AI-generated text to examine how technology shapes self-perception. Includes the full text of "Ghosts" alongside new experiments feeding chapters into ChatGPT for feedback and incorporating crowdsourced responses.
 - **Relevance to demo:** Establishes the broader intellectual context for using AI to interrogate selfhood. The crowdsourced-response chapter in particular parallels your project's interest in how different readers' contexts produce different encounters with the same emotional architecture.
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VI. THE INTERACTIVE NARRATIVE LANDSCAPE: What Exists and What Doesn't

These sources map the state of interactive/adaptive narrative — establishing that your specific combination (fixed architecture, reader-context-populated threshold, real-time current events) is genuinely novel.

10. Kreminski, M., Dickinson, M., & Wardrip-Fruin, N. (2025). WhatELSE: Shaping narrative spaces at configurable level of abstraction for AI-bridged interactive storytelling. *arXiv:2502.18641*.

- **URL:** <https://arxiv.org/pdf/2502.18641>
- **Key contribution:** An authoring system that creates narrative possibility spaces from example stories, using linguistic abstraction to control boundaries. Provides three views (narrative pivot, outline, variants) for authors to understand and shape the space of possible narratives.
- **Relevance to demo:** The most architecturally sophisticated academic system for constrained generative narrative. But it personalizes based on *player choices within the story*, not *reader context from outside the story*. Your project occupies a gap this system doesn't fill.

11. *Bury Me, My Love* (2017). Developed by The Pixel Hunt and Figs. Published by ARTE.

- **URL:** <https://burymemylove.arte.tv/>
- **Description:** A mobile game about a Syrian refugee's journey to Europe, played through a simulated messaging interface in real time. When the character says she'll text in the morning, the player waits until the next day. Based on the real WhatsApp messages of Dana, a Syrian woman, reported in *Le Monde*.
- **Relevance to demo:** The closest existing work to Subtext's temporal personalization. Uses the reader's actual time and phone as narrative material. But it doesn't use disclosed personal data or current events; its personalization is purely temporal and choice-based.

12. *Black Mirror: Bandersnatch* (2018). Directed by David Slade. Netflix.

- **Key data point:** Netflix confirmed it saved and stored every viewer choice against individual accounts to "help determine how to improve this model of storytelling" and "inform personalized recommendations." A GDPR request revealed Netflix stored choice data, timestamps, platform, and rewatch status.
 - **Relevance to demo:** The highest-profile case of interactive narrative as data collection. But *Bandersnatch* collects *narrative preferences* (Frosties or Sugar Puffs). Your project collects *phenomenological self-disclosure* (what does the light look like). That's a qualitatively different kind of data with different implications for intimacy and power.
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VII. THE INDUSTRIAL CONTEXT: Personalization as Identity Construction

These sources document how existing personalization systems construct user identity and manufacture the sensation of being known — the commercial baseline your project critiques.

13. Spotify: Wrapped and Personalization Infrastructure

- **Key reference:** Spotify patented recommendation technology capable of tailoring music based on emotional state, gender, age, and accent — inferring mood through speech, ambient cues, and usage patterns. Wrapped transforms listening data into shareable identity artifacts. Over 156 million users interacted with Wrapped in 2023.
- **Relevance to demo:** Wrapped is personalization-as-flattery: it mirrors you back to yourself in a way that feels affirming and shareable. Your project is personalization-as-ambush: it uses your data to make you feel something uncomfortable. The contrast between these two modes is the demo's argument about power.

14. Netflix Recommendation System

- **Key data point:** Netflix reports that 80% of streaming hours come from homepage recommendations rather than search. The recommendation algorithm analyzes viewing history, habits of similar users, behavior patterns, and time-of-day viewing preferences.
 - **Relevance to demo:** Establishes the scale at which personalization already governs attention. Your project makes the mechanism visible; Netflix keeps it invisible. That difference is the difference between a literary instrument and an engagement engine.
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VIII. SUPPLEMENTARY: Persuasion Theory and AI Ethics

15. Teeny, J. D. & Matz, S. C. (2024). We need to understand "when" not "if" generative AI can enhance personalized persuasion. *Proceedings of the National Academy of Sciences*, 121(43), e2418005121.

- **DOI:** 10.1073/pnas.2418005121
- **URL:** <https://pmc.ncbi.nlm.nih.gov/articles/PMC11513937/>
- **Key argument:** Response to Hackenburg & Margetts (2024), who found that demographic-based LLM personalization didn't increase persuasion. Teeny & Matz argue the question is not *whether* AI can personalize persuasion but *when* — i.e., which psychological dimensions, prompting strategies, and contexts produce the effect. The debate itself is relevant: it shows the field is actively mapping the conditions under which personalized generation becomes influence.
- **Relevance to demo:** Your project is an empirical contribution to this debate: you're testing whether phenomenological context (time, light, social position) is a dimension that produces the "recognition" effect.

16. Hackenburg, K. & Margetts, H. (2024). Evaluating the persuasive influence of political microtargeting with large language models. *Proceedings of the National Academy of Sciences*, 121(43), e2403116121.

- **DOI:** 10.1073/pnas.2403116121
- **Key finding:** Using an LLM to personalize political messages based on demographic attributes did not make messages more persuasive in a controlled experiment. Authors note this tests only one LLM and one prompting approach — not a refutation of personalized persuasion broadly.
- **Relevance to demo:** Important counterpoint. Demonstrates that personalization isn't magic — the *dimension* of personalization matters. Your project personalizes on phenomenological experience (time of day, quality of light, social context), not demographics. This distinction is where your contribution lies.

CITATION FORMAT FOR DEMO SLIDE

For the final slide of your deck, a compressed version:

Research Foundation

Matz et al. (2024), *Scientific Reports* — LLM-personalized messages are significantly more persuasive from a single psychological prompt

Peters & Matz (2024), *PNAS Nexus* — LLMs infer Big Five traits from social media posts at supervised-ML accuracy

Kosinski, Stillwell & Graepel (2013), *PNAS* — Facebook Likes predict personality, orientation, politics from minimal data

Matz et al. (2017), *PNAS* — Psychologically targeted ads are up to 63% more effective in field experiments (N = 3.5M)

De Freitas et al. (2025), *HBS Working Paper* — AI companions deploy emotional manipulation at scale; 43% of farewells use dark patterns

Vara (2021), "Ghosts," *The Believer* — AI-assisted literary writing as instrument, not shortcut; seams as meaning

This corpus was assembled for the Subtext demo for BUSGEN 116: Free Systems. It is organized to support the demo's argument: that the same personalization infrastructure powering recommendation engines, targeted advertising, and AI companions can be turned inside out — made experiential, made visible, and made to ask something of the reader rather than extract something from them.